

Topics and how they change over time

```
tx <- list(
  seed = 2026L,
  min_docfreq = 5L, # drop words in fewer paragraphs than this
  run_searchK = FALSE, # TRUE once to choose K (slow)
  K_grid = seq(5, 30, 3),
  K = 12L, # set from the searchK plots
  spline_df = 5L, # flexibility of the year trend
  run_sensitivity = FALSE, # TRUE once to test the merge settings
  sens_factors = c(0.75, 1.25), # multiples of min_tokens and max_tokens
  n_words = 20L, # words per topic shown to the codebook model
  n_examples = 5L, # example paragraphs per topic shown to the codebook model
  n_draws = 25L, # draws of topic shares behind each trend interval
  workers = 8L,
  use_openrouter = TRUE, # FALSE at work
  steps = tibble::tribble(
    ~step, ~base_url_env, ~api_key_env, ~model,
    "codebook", "COMPASS_LARGE_URL", "COMPASS_LARGE_KEY", "LARGE-MODEL")
  suppressPackageStartupMessages({
    library(here); library(stm); library(splines); library(viridis)
    library(tidyverse) # tidyverse last so its verbs win
  })
  theme_set(theme_minimal(base_size = 12))
  set.seed(tx$seed)
  source(here::here("code", "llm_source.R"))
  if (isTRUE(tx$use_openrouter)) {
    source(here::here("code", "llm_openrouter.R"))
    tx$steps$model <- open_router_models[tx$steps$step]
  }
}
```

Input

```
para_in <- p_out("paragraphs.csv")
paragraphs <- readr::read_csv(para_in, show_col_types = FALSE)
writeLines(glue::glue("{nrow(paragraphs)} paragraph units from \\
  {dplyr::n_distinct(paragraphs$atom_id)} comments."))
```

951 paragraph units from 60 comments.

Document-term matrix

Words are lowercased and kept only if purely alphabetic, which drops numbers and special characters. Stopwords and month and weekday names are removed; temporal words would let topics encode the year directly and leak time into the trends. Words in fewer than 5 paragraphs are dropped.

```
tokens <- clean_tokens(paragraphs)
stm_in <- stm_input(tokens, tx$min_docfreq)
# stm needs a plain data frame (not a tibble) in the same row order as the
# documents, and the year spline as ordinary columns yr1, yr2, ...
meta <- tibble(para_uid = names(stm_in$documents)) |>
  dplyr::left_join(paragraphs, by = "para_uid") |>
  as.data.frame()
bs <- splines::ns(meta$Year, df = tx$spline_df)
colnames(bs) <- str_c("yr", seq_len(ncol(bs)))
meta <- cbind(meta, as.data.frame(bs))
prev_rhs <- str_c(colnames(bs), collapse = " + ")
prev_form <- as.formula(str_c("~ ", prev_rhs))
glue::glue("{length(stm_in$documents)} paragraphs, {length(stm_in$vocab)}
words")
```

951 paragraphs, 2517 words

Choosing K

Take the smallest K where held-out likelihood has flattened and residual dispersion has stopped falling, then break ties on coherence and exclusivity. If the best values sit at the edge of the grid, widen the grid.

```
if (isTRUE(tx$run_searchK)) {
  future::plan(future::multisession, workers = tx$workers)
  ho <- stm::make_heldout(stm_in$documents, stm_in$vocab, seed = tx$seed)
  metrics <- cached(p_out("searchK.rds"), para_in,
    furrr::future_map(tx$K_grid, function(k) {
      fit <- stm::stm(ho$documents, ho$vocab, K = k, init.type = "Spectral",
        prevalence = prev_form, data = meta, verbose = FALSE)
      tibble(K = k,
        heldout = stm::eval_heldout(fit, ho$missing)$expected_heldout,
        dispersion = stm::checkResiduals(fit, ho$documents)$dispersion,
        coherence = mean(stm::semanticCoherence(fit, ho$documents)),
        exclusivity = mean(stm::exclusivity(fit)))
    }, .options = furrr::furrr_options(seed = tx$seed)) |> purrr::list_rbind()
  future::plan(future::sequential)
  metrics |>
    tidyr::pivot_longer(-K) |>
    ggplot(aes(K, value)) +
```

```
geom_line(color = viridis(1, begin = 0.35)) + geom_point() +
  facet_wrap(~ name, scales = "free_y") + labs(y = NULL)
} else glue::glue("searchK skipped; K = {tx$K}")
```

```
searchK skipped; K = 12
```

Structural topic model

Year enters topic prevalence as a spline, so each topic's share can rise and fall over the period. Spectral initialization makes the fit reproducible.

```
stm_fit <- cached(p_out("stm_fit.rds"), para_in,
  stm::stm(stm_in$documents, stm_in$vocab, K = tx$K, init.type = "Spectral",
    prevalence = prev_form, data = meta, verbose = FALSE))
K <- ncol(stm_fit$theta)
if (K != tx$K) stop("stm_fit.rds has K = ", K, " but tx$K = ", tx$K,
  ". Delete stm_fit.rds, then re-render.")
theta <- stm_fit$theta
colnames(theta) <- str_c("topic_", seq_len(K))
```

Assigning paragraphs to topics

Each paragraph is assigned its most likely topic. **Entropy R^2** (0 to 1) says how sharply paragraphs sort into topics overall; **AvePP** is the average topic share among a topic's own paragraphs, with about 0.7 the usual floor. Both fall as units get longer, because longer units genuinely mix topics, so use them to compare topics within a fit, not to choose the unit size.

```
paragraph_theta <- meta |>
  dplyr::select(para_uid, atom_id, para_id, Year, n_tokens, text) |>
  dplyr::bind_cols(tibble::as_tibble(theta)) |>
  dplyr::mutate(topic = max.col(theta, ties.method = "first"),
    theta_max = theta[cbind(dplyr::row_number(), topic)])
entropy_r2 <- 1 - (-sum(theta * log(theta + 1e-12))) / (nrow(theta) * log(K))

words <- stm::labelTopics(stm_fit, n = tx$n_words)
topic_summary <- tibble(
  topic = seq_len(K),
  prevalence = colMeans(theta),
  frex = apply(words$frex, 1, str_c, collapse = ", "),
  prob = apply(words$prob, 1, str_c, collapse = ", ")) |>
  dplyr::left_join(paragraph_theta |>
    dplyr::group_by(topic) |>
    dplyr::summarise(n paras = dplyr::n(), avepp = mean(theta_max)),
    by = "topic")
```

```

readr::write_csv(paragraph_theta, p_out("paragraph_theta.csv"))
readr::write_csv(topic_summary, p_out("topic_summary.csv"))
theta_in <- p_out("paragraph_theta.csv")

writeLines(glue::glue("Entropy R2 = {round(entropy_r2, 3)}. \\
  Topics with AvePP under 0.7: {sum(topic_summary$avepp < 0.7, na.rm = TRUE)}
of {K}."))

```

Entropy R2 = 0.591. Topics with AvePP under 0.7: 11 of 12.

```

topic_summary |>
  dplyr::arrange(dplyr::desc(prevalence)) |>
  dplyr::select(topic, prevalence, n paras, avepp) |>
  tbl_paged(caption = "Topics by prevalence.")

```

Topics by prevalence.

topic	prevalence	n paras	avepp
7	0.136	142	0.655
10	0.100	104	0.647
5	0.092	80	0.647
6	0.089	79	0.711
4	0.087	91	0.633
2	0.079	81	0.629
8	0.075	76	0.592
9	0.074	60	0.694
3	0.072	69	0.609
1	0.069	58	0.630
12	0.068	62	0.642
11	0.058	49	0.652

Do the merge settings drive the topics?

The paragraph units are rebuilt at 0.75 and 1.25 times `min_tokens` and `max_tokens`, from the embeddings already saved (no API calls), and the model is refit at the same `K`. Each baseline topic is matched to its closest topic in each refit by the cosine of their word distributions. High cosines mean the topics belong to the corpus; a topic with a low cosine exists only at one way of cutting the text and deserves caution.

```

if (isTRUE(tx$run_sensitivity)) {
  sens_in <- p_out(c("paragraphs.csv", "blocks.csv", "embeddings.rds",
                    "merge_settings.rds"))
  blocks <- readr::read_csv(p_out("blocks.csv"), show_col_types = FALSE)
  emb <- readRDS(p_out("embeddings.rds"))
  ms <- readRDS(p_out("merge_settings.rds"))
  refit <- function(f) {
    p <- link_corpus(blocks, emb$vec, ms$min_tokens * f, ms$max_tokens * f,
                     ms$min_sim) |>
    dplyr::filter(keep)
    s <- stm_input(clean_tokens(p), tx$min_docfreq)
    list(n = length(s$documents),
         fit = stm::stm(s$documents, s$vocab, K = K, init.type = "Spectral",
                        verbose = FALSE))
  }
  sens <- cached(p_out("sensitivity_fits.rds"), sens_in,
                 purrr::map(tx$sens_factors, refit) |>
                 purrr::set_names(tx$sens_factors))
  sens_tbl <- purrr::imap(sens, function(x, f)
    match_topics(beta_matrix(stm_fit), beta_matrix(x$fit)) |>
    dplyr::mutate(factor = as.numeric(f), paragraphs = x$n) |>
    purrr::list_rbind())
  readr::write_csv(sens_tbl, p_out("sensitivity.csv"))
  sens_tbl |>
    dplyr::group_by(factor) |>
    dplyr::summarise(paragraphs = dplyr::first(paragraphs),
                     median_cosine = stats::median(cosine),
                     lowest_cosine = min(cosine)) |>
    tbl_paged(caption = "Matched topic cosine against the baseline fit.")
  sens_tbl |>
    tidyr::pivot_wider(id_cols = topic, names_from = factor,
                       values_from = cosine, names_prefix = "x") |>
    tbl_paged(caption = "Matched cosine per topic.")
} else writeLines("Sensitivity test skipped (tx$run_sensitivity = FALSE).")

```

Sensitivity test skipped (tx\$run_sensitivity = FALSE).

Topic codebook

One call per topic, all in parallel, with the larger model. Each call sees only that topic's evidence: its 20 most distinctive (FREX) and 20 most probable words and its 5 most typical paragraphs. It drafts a label, a one-sentence description, and a candidate proposition for stance work. The file is written once: if `topic_codebook.csv` exists it is read, not redrafted, so your edits survive. Delete it to redraft.

```

cb_path <- p_out("topic_codebook.csv")
check_inputs(cb_path, theta_in)

```

```

if (file.exists(cb_path)) {
  codebook <- readr::read_csv(cb_path, show_col_types = FALSE)
  writeLines(glue::glue("Existing codebook read ({nrow(codebook)} topics); no
calls made. "))
} else {
  examples <- paragraph_theta |>
    dplyr::group_by(topic) |>
    dplyr::slice_max(theta_max, n = tx$n_examples, with_ties = FALSE) |>
    dplyr::summarise(ex = str_c("-", str_trunc(text, 400), collapse = "\n"))
  ev <- dplyr::left_join(topic_summary, examples, by = "topic")
  sys <- build_prompt(
    role = "You are a survey methodologist who names statistical topics so
colleagues can use them.",
    task = str_c("Read one topic's words and example paragraphs from public
comments. Write a ",
                  "short label, one factual sentence on what the paragraphs
discuss, and one ",
                  "proposition: a single claim, in one sentence, that a paragraph
on this ",
                  "topic could agree or disagree with. Check every word against
the evidence."),
    rules = c("Use only the words and paragraphs shown.",
              "Do not infer who is speaking or their intent.",
              "Do not invent statistics, dates, places, or names.",
              "The proposition states one side only: never 'X over Y' or 'X
rather than Y'."),
    output = "A label of at most 4 words, a one-sentence description, and a one-
sentence proposition.")
  type <- ellmer::type_object(
    label = ellmer::type_string("At most 4 words."),
    description = ellmer::type_string("One factual sentence."),
    proposition = ellmer::type_string("One claim, one sentence. "))
  prompts <- str_c("Distinctive words: ", ev$frex, "\nMost probable words: ",
ev$prob,
                  "\nExample paragraphs:\n", ev$ex)
  drafts <- ask_many(make_chat("codebook", sys, tx$steps, tx$use_openrouter),
                    prompts, type)
  codebook <- tibble(topic = ev$topic, label = drafts$label,
                    description = drafts$description,
                    proposition = drafts$proposition) |>
    dplyr::arrange(topic)
  readr::write_csv(codebook, cb_path, na = "")
  record_inputs(cb_path, theta_in)
  if (anyNA(codebook$label))
    warning(sum(is.na(codebook$label)), " topics got no label; name them by
hand.")
  writeLines(glue::glue("Codebook drafted for {nrow(codebook)} topics. Edit \\

```

```

    topic_codebook.csv; it is not redrafted while it exists."))
}

```

Codebook drafted for 12 topics. Edit topic_codebook.csv; it is not redrafted while it exists.

```

dup <- codebook |> dplyr::add_count(label) |> dplyr::filter(n > 1, !is.na(label))
if (nrow(dup)) writeLines(glue::glue("***Duplicate labels to fix:** \\
  {str_c('topic ', dup$topic, ' (', dup$label, ')', collapse = ', ')}."))

codebook |>
  dplyr::left_join(dplyr::select(topic_summary, topic, frex), by = "topic") |>
  dplyr::mutate(title = str_c("Topic ", topic, ": ", label)) |>
  tbl_records("title", c("description", "proposition", "frex"),
    caption = "Topic codebook. Edit the CSV; nothing redrafts it.")

```

Topic codebook. Edit the CSV; nothing redrafts it.

Topic 1: Civil War Context - description: The paragraphs discuss the context and causes of a civil war, the state of the nation, and the role of faith and providence in resolving differences. - proposition: The nation's security depends on its acknowledged dependence upon God and His overruling providence. - frex: let, differences, religious, around, divide, earth, capacity, faculties, british, physical, surely, war, fathers, rest, prevail, unite, still, five, present, man

Topic 2: National Defense Commerce - description: The paragraphs discuss the relationship between national defense, commerce, and the government's role in protecting and promoting the country's economy and security. - proposition: The government should prioritize protecting national commerce and industry through defense measures and policy. - frex: revenue, currency, army, debt, trade, navy, credit, south, naval, coast, commerce, necessary, industry, tariff, manufactures, invasion, agriculture, taxes, products, pay

Topic 3: Inauguration Gratitude - description: The paragraphs discuss the sentiments and acknowledgments expressed by an individual upon taking office. - proposition: The person taking office feels adequately prepared to discharge their duties. - frex: discharge, gratitude, duties, station, countrymen, solemn, almighty, presence, blessed, sustain, enter, trust, generous, occasion, hopes, beloved, high, providence, devotion, arduous

Topic 4: Federal Power Limits - description: The paragraphs discuss the distribution and limits of power among different levels of government in the United States. - proposition: The Constitution clearly defines the limits of power for the general government and the states. - frex: union, territory, confederacy, constitution, granted, territories, general, grant, sovereignty, settled, powers, states, exercised, domestic, texas, authority, constitutional, blessings, separate, institutions

Topic 5: Government and Economy - description: The paragraphs discuss the role of government and economy in ensuring the well-being of citizens, particularly the vulnerable, and addressing issues like poverty and inequality. - proposition: A great nation must care for the vulnerable and protect its people from life's worst hazards and misfortunes. - frex: women, work, families,

children, health, wealth, lives, something, know, economy, society, succeed, built, forces, schools, workers, land, child, rich, cost

Topic 6: Global Unity and Freedom - description: The paragraphs discuss the importance of unity and cooperation among free nations and peoples in achieving peace, freedom, and prosperity. - proposition: United, free nations can effectively safeguard their spiritual and cultural heritage. - frex: peoples, seek, weapons, help, program, peace, nations, helped, world, threat, freedom, worlds, nuclear, strive, strong, enemies, free, price, areas, strength

Topic 7: American Renewal Story - description: The paragraphs discuss the themes of American renewal, unity, and the country's story of progress and challenges. - proposition: America's story is one of continuous progress towards a more perfect union. - frex: americas, america, promise, heart, americans, journey, story, generation, let, century, dignity, together, generations, challenges, forward, voices, dreams, us, challenge, resolve

Topic 8: America's Global Role - description: The paragraphs discuss America's values, its role in the world, and its relationships with other nations. - proposition: America should prioritize its relationships with other nations to promote peace and justice. - frex: self, civilization, thought, ideals, understanding, problems, republic, economic, vision, foundations, wish, fail, lies, aided, realization, profit, mankind, life, courage, counsel

Topic 9: Foreign Policy Principles - description: The paragraphs discuss the principles and conduct of a nation's foreign policy and its relations with other nations. - proposition: A nation's foreign policy should prioritize cultivating friendly relations with all other nations. - frex: relations, foreign, policy, press, rights, suffrage, friendly, respect, settlement, nations, friendship, universal, advantage, obligations, cultivate, ballot, arbitration, intercourse, controversies, ignorance

Topic 10: Government Reform - description: The paragraphs discuss the need for reform in various aspects of government, including civil service, law enforcement, and administration. - proposition: The government should adopt a merit-based system for appointments to office. - frex: business, enforce, laws, amendment, enforcement, legislation, partisan, expenditures, labor, interstate, due, money, public, session, official, combinations, taxation, reform, popular, election

Topic 11: Separation of Powers - description: The paragraphs discuss the distribution and exercise of power among the different branches of government. - proposition: The Constitution intended for the Executive branch to play a significant role in the legislative process. - frex: measures, legislature, executive, distinguished, former, conduct, heretofore, causes, enlightened, certainly, contest, occurred, department, instance, early, circumstances, formed, least, although, opinion

Topic 12: Presidential Inauguration Speeches - description: The paragraphs discuss inauguration ceremonies of US presidents. - proposition: Inauguration ceremonies celebrate the unity of the country. - frex: oath, president, thank, vice, day, mr, ago, going, taken, ceremony, dont, years, four, today, washington, inauguration, celebrate, chief, first, accept

Prevalence over time

For each topic, its share of each paragraph is regressed on the year spline with a fractional logit (a quasibinomial model for proportions; Papke and Wooldridge 1996), so every estimated share stays between 0 and 1. This replaces `stm::estimateEffect`, whose linear model can predict negative shares for rare topics. The model is refit on 25 draws of the topic shares from the fitted STM and the results are combined, so the 95 percent intervals include the topic model's own uncertainty as well as the fact that paragraphs from one comment are related.

```
draws <- withr::with_seed(tx$seed, purrr::map(seq_len(tx$n_draws), function(i)
  do.call(rbind, stm::thetaPosterior(stm_fit, nsims = 1, type = "Global"))))
grid <- sort(unique(meta$Year))
topic_trends <- purrr::map(seq_len(K), function(k)
  topic_trend(draws, k, bs, grid, meta$atom_id)) |>
  purrr::list_rbind()
readr::write_csv(topic_trends, p_out("topic_trends.csv"))
```

```
pages <- topic_trends |>
  dplyr::left_join(dplyr::select(codebook, topic, label), by = "topic") |>
  dplyr::left_join(dplyr::select(topic_summary, topic, prevalence), by =
    "topic") |>
  dplyr::mutate(name = str_c(topic, ". ", str_trunc(label, 24)),
    page = (dplyr::dense_rank(dplyr::desc(prevalence)) - 1) %/% 9)
|>
  dplyr::group_split(page)
purrr::walk(pages, function(d) print(
  ggplot(d, aes(Year, fit)) +
    geom_ribbon(aes(ymin = lo, ymax = hi), alpha = 0.2, fill = viridis(1, begin
    = 0.35)) +
    geom_line(color = viridis(1, begin = 0.35)) +
    facet_wrap(~ name, scales = "free_y", ncol = 3) +
    labs(x = NULL, y = "expected topic share")))
```

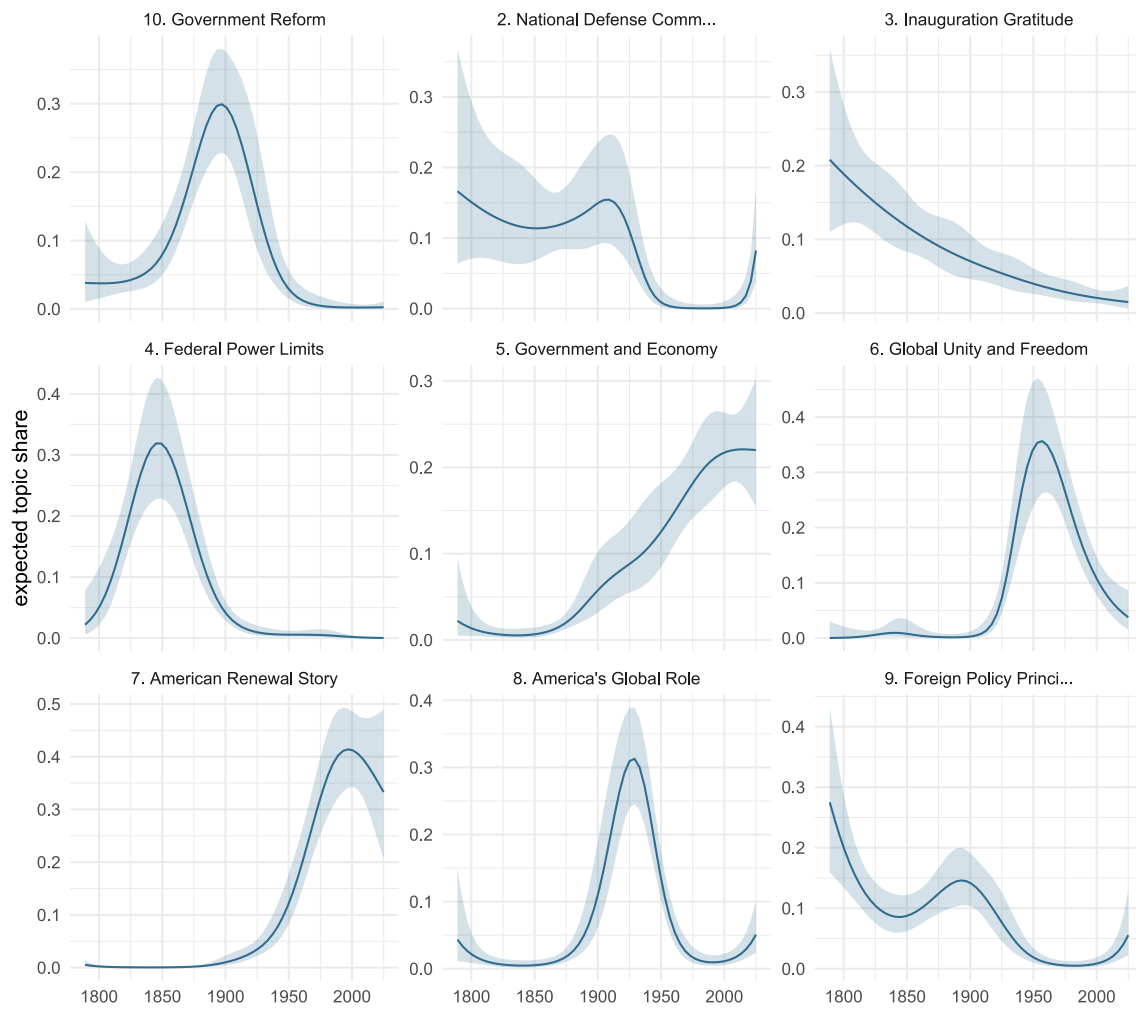


Figure 1: Expected topic share by year with 95 percent intervals, nine topics per page, most prevalent first.

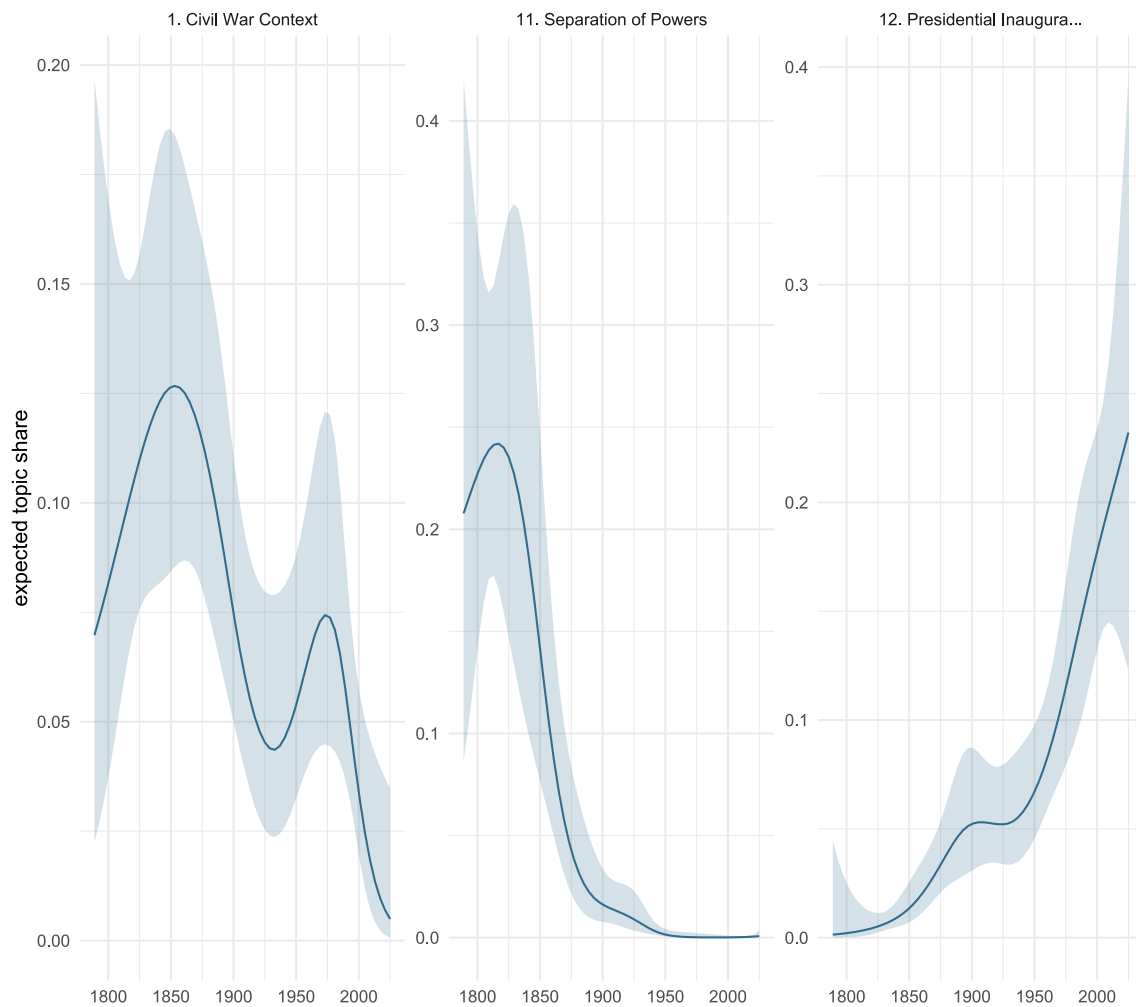


Figure 2: Expected topic share by year with 95 percent intervals, nine topics per page, most prevalent first.

```
writeln(glue::glue("Wrote paragraph_theta.csv, topic_summary.csv, \\
  topic_codebook.csv, topic_trends.csv, stm_fit.rds. \\
  The app reads the four CSVs from outputs/."))
```

Wrote paragraph_theta.csv, topic_summary.csv, topic_codebook.csv, topic_trends.csv, stm_fit.rds. The app reads the four CSVs from outputs/.